UNIT-II

1. Introduction

Animation means making pictures move. It is done by showing many still images very quickly, which looks like real movement. Multimedia means using many forms like text, pictures, sound, and video together.

Key Points about Animation in Multimedia

- **Illusion of Motion**: Animation shows a series of slightly different images very fast, which tricks our eyes into thinking the object is moving.
- **2D and 3D Animation**
  2D animation uses flat images (like cartoons).
  3D animation makes objects look real with depth and volume (like animated movies and games).
- **Different Animation Techniques**
  Examples include:
  - Traditional hand-drawn animation (cel animation)
  - Stop-motion (moving physical objects frame by frame)
  - Computer-generated animation (CGI)
- **Mixing with Multimedia**: Animation can be combined with text, sound, and video to create more exciting and interactive content.
- **Uses in Many Fields**: Animation is widely used in:
  - **Movies & TV shows** – bringing characters and stories to life
  - **Video games** – showing movements of characters and objects
  - **Marketing & Ads** – explainer videos, animated logos, product ads
  - **Education** – making lessons fun and explaining difficult ideas visually

2. History of Animation in Multimedia

The history of animation is very old. It started with simple drawings and devices, and today we have advanced computer-generated imagery (CGI). Over time, animation has grown from cave paintings to 3D movies, video games, and mobile apps.

Early Forms and Pioneers

- **Prehistoric and Ancient Times**
  Cave paintings and Egyptian art tried to show movement by drawing the same figure in different positions.
- **Magic Lantern (1603)**
  An early projector that used light and glass slides to show moving pictures.
- **Phenakistoscope (1832)**
  Created by Joseph Plateau and Simon Stampfer, it showed moving images when spun, using the "persistence of vision" principle.
- **Early Film Pioneers**
  Eadweard Muybridge showed motion through photographs of running horses.
  J. Stuart Blackton made some of the first animated films.
- **Émile Cohl (1908)**
  Known as the "Father of Animation," he created *Fantasmagorie*, one of the first hand-drawn animations.
- **Winsor McCay (1914)**
  Made *Gertie the Dinosaur*, introducing techniques like **keyframes** and **loops**, still used in animation.

Traditional Animation (2D)

- **Cel Animation**
  Drawings were made on transparent sheets (cels) and photographed one by one to create motion.
- **Disney's Success**
  Movies like *Snow White*, *Pinocchio*, and *Bambi* made Disney famous and set the standard for animation.

- **Limited Animation**
  Used in TV shows like *The Flintstones* and *Yogi Bear*. Fewer drawings were made to save time and money.

Computer Animation

- **Early Computer Animation (1960s)**
  Movies like *Catalogue (1961)* and *Computer Ballet (1965)* showed the first digital animations.
- **John Whitney**
  Called one of the fathers of computer animation. He used machines and created experimental animations.
- **3D Animation and CGI**
  Pixar's *Toy Story (1995)* was the first fully computer-animated movie, changing the animation industry.
- **Motion Graphics**
  Animated text and graphics used in film titles, ads, and websites.

Multimedia Integration

- **Television**
  Cartoons became popular on TV, both for kids and adults.
- **Video Games**
  Animation is used to design characters, actions, and game worlds.
- **Websites and Apps**
  Animations make apps and websites more engaging and easier to use.

3. Uses of Animation in Multimedia

Animation makes content more interesting, easy to understand, and engaging. It is used to explain ideas, tell stories, and make learning, marketing, and entertainment more fun.

1. Storytelling and Engagement

- Makes stories simple and easy to follow.
- Grabs attention with interesting visuals.
- Brings characters and imaginary worlds to life.

2. Education and Training

- Explains difficult ideas clearly.
- Simulates real-life situations for practice.
- Helps people remember information better.

3. Marketing and Advertising

- Creates eye-catching ads and videos.
- Promotes products, brands, and services in creative ways.
- Popular on social media through animated videos and GIFs.

4. Entertainment

- Used in movies, TV shows, and video games.
- Creates exciting characters and imaginary worlds.
- Can make music videos, short films, and fun content.

5. Data Visualization and Communication

- Shows complex data in simple visuals.
- Animated charts and graphs explain information clearly.
- Helps people understand and remember data better.

6. Other Applications

- **Medical:** Shows surgeries and anatomy clearly.
- **Engineering & Architecture:** Visualizes designs and processes.
- **UI Design:** Makes websites and apps interactive.
- **Gaming:** Creates characters, worlds, and special effects.
- **Social media:** Makes posts and stories more engaging.
- **Motion Graphics:** Animates titles, credits, and graphics.

4. Types of Animation in Multimedia

Animation in multimedia comes in different types, each with its own style and uses.

1. 2D Animation

- Drawings or images in two dimensions (flat).
- Used in cartoons, animated series, and explainer videos.

2. 3D Animation

- Creates three-dimensional images using computers.
- Looks realistic and is used in movies, video games, and product demos.

3. Traditional Animation (Cel Animation)

- Each frame is drawn by hand (or digitally now).
- Old cartoons and classic animated movies use this.

4. Stop Motion Animation

- Uses real objects moved little by little and photographed.
- Examples: Claymation (clay figures) and cutout animation (flat materials).

5. Motion Graphics

- Animates text, shapes, and graphics.
- Common in titles, infographics, and ads.

6. Rotoscoping

- Tracing over real video frames to make animation or add effects.
- Creates a stylized or special visual effect.

7. Motion Capture

- Records movements of real people or objects.
- Transfers those movements to digital characters for realistic animation.

5. Principles of Animation

Animation is made by showing many images quickly to create the illusion of movement. To make animation realistic and interesting, there are **12 key principles**:

1. Squash and Stretch

- Shows how objects change shape when they move.
- Makes animation look realistic.
- *Example:* A bouncing ball squashes when it hits the ground.

2. Anticipation

- Prepares the audience for an action.
- Makes movement look real.
- *Example:* A batsman raises the bat before hitting the ball.

3. Arcs

- Most movements happen in curved paths, not straight lines.
- *Example:* The swing of a bowler's hand follows an arc.

4. Slow In – Slow Out

- Objects take time to speed up or slow down.
- *Example:* A car gradually accelerates and decelerates.

5. Appeal

- Characters and visuals should be attractive and easy to understand.
- Avoid awkward designs or confusing shapes.

6. Timing

- The speed of movement affects the feel of the animation.
- *Example:* Fast movement = energetic, slow movement = lazy or calm.

7. 3D Effect

- Adding depth makes objects more realistic.

- *Example:* A cube looks 3D, a square is flat.

8. Exaggeration

- Emotions or actions are slightly overdone to make them clear and fun.
- Keep balance if multiple elements are present.

9. Staging

- Show the main idea or action clearly.
- Avoid unnecessary details.
- *Example:* Focus on the main action a character is doing.

10. Secondary Action

- Smaller movements that support the main action.
- *Example:* While drinking tea, hand movements and facial expressions are secondary actions.

11. Follow Through

- Movements continue even after the main action ends.
- *Example:* After throwing a ball, the hand keeps moving.

12. Overlap

- One action starts before another finishes.
- *Example:* Drinking tea with the right hand while the left hand reaches for a sandwich.

6. Various Techniques of Animation

1. Traditional Animation (Hand-Drawn / Cel Animation)

- Draw each frame by hand on paper or transparent sheets (cels).
- Color and photograph each frame against backgrounds.
- Modern version uses computers for coloring and effects.
- **Full Animation:** Smooth, detailed (e.g., Disney movies).
- **Limited Animation:** Fewer frames, cheaper and faster (e.g., TV cartoons).
- **Rotoscoping:** Tracing live-action footage.
- **Live-action Blending:** Mixing animated characters with real actors.

2. Stop Motion Animation

- Real objects moved slightly and photographed frame by frame.
- **Types:**
  - **Puppets:** Movable joint figures (*The Nightmare Before Christmas*).
  - **Puppetoons:** Different puppet versions for frames (George Pal).
  - **Clay Animation / Claymation:** Clay figures (*Wallace & Gromit*, *Chicken Run*).
  - **Strata-cut:** Clay loaf sliced to reveal movement.
  - **Cutout Animation:** Paper or cloth pieces moved frame by frame (*South Park*).
  - **Silhouette Animation:** Characters appear as dark shapes (*The Adventures of Prince Achmed*).
  - **Model Animation:** Stop-motion in live-action world (*King Kong*).
  - **Go Motion:** Adds motion blur (*Star Wars: Empire Strikes Back*).
  - **Object Animation:** Everyday objects animated.
  - **Graphic Animation:** Photos or clippings animated.
  - **Brickfilm:** LEGO/brick-based animation.
  - **Pixilation:** Humans as stop-motion characters.

3. Computer Animation

- Created digitally on computers.
- **2D Animation:** Flat images animated digitally (Flash, After Effects, PowerPoint).
- **3D Animation:**
  - 3D models manipulated with digital skeletons.
  - Includes physics, hair, fire, water effects.
  - **Cel Shading:** 3D looks like 2D cartoon (*The Iron Giant*).

- **Machinima:** Animation using video game engines.
- **Motion Capture:** Record human movements for digital characters (*Polar Express*).
- **Physically Based Animation:** Realistic movement using computer simulations.

4. Mechanical Animation

- **Animatronics / Audio-Animatronics:** Robots that move and make sound (Disney).
- **Linear Animation Generator:** Motion illusion using static images.
- **Chuckimation:** Objects thrown or moved to create motion (*Action League Now!*).
- **Magic Lantern:** Early projected moving images (1600s).

5. Other Special Techniques

- **World of Color / Musical Fountain:** Water, lights, and projections create animation.
- **Drawn-on-Film Animation:** Draw directly on film (Norman McLaren).
- **Paint-on-Glass:** Manipulate oil paints on glass (Aleksandr Petrov).
- **Erasure Animation:** Draw and erase images to animate (William Kentridge).
- **Pinscreen Animation:** Pins create textures and shadows.
- **Sand Animation:** Sand moved on glass to animate.
- **Flip Book:** Pages with images flip to show motion.
- **2.5D Animation:** Mix of 2D and 3D for depth illusion.

7. Animation On The Web

Web animation uses moving visuals on websites to make them attractive, interactive, and easy to understand. It improves user experience and helps communicate ideas effectively.

Key Points

1. **Visual Appeal & Engagement:**
   - Animations make websites look lively and interesting.
   - They grab attention and encourage users to explore more.
2. **Better User Experience:**
   - Guides users through the site.
   - Highlights important things.
   - Shows feedback for actions (like clicking buttons).
3. **Effective Communication:**
   - Explains ideas or product features clearly.
   - Helps tell stories in a simple way.
4. **Types of Web Animation:**
   - **GIFs:** Small moving images.
   - **CSS Animations:** Styling animations using code.
   - **JavaScript Animations:** More advanced interactive animations.
   - **WebGL:** 3D animations on websites.

- **Integration with Multimedia:** Combined with text, images, audio, and video for interactive experiences.

Examples

- **Micro-interactions:** Button hover effects, loading animations.
- **Animated Logos:** Rotating or moving logos for branding.
- **Interactive Tutorials:** Step-by-step guides with animation.
- **Animated Infographics:** Data shown in a fun, easy-to-understand way.
- **Full-screen Animations:** Large visuals to create immersive experiences.

8. 3D Animation

3D animation is the process of creating characters, objects, and environments in a 3-dimensional space to make them look real and move naturally. It is done using special 3D software.

Key Steps in 3D Animation

1. **3D Modeling:** Create the shapes and structure of objects or characters.
2. **Texturing:** Add colors, patterns, and surface details to the models.
3. **Rigging:** Build a skeleton for characters so they can move.
4. **Animation:** Make the characters and objects move and act.
5. **Rendering:** Produce the final images or video with lighting and shadows.
6. **Compositing:** Combine the animation with backgrounds, text, or other effects.

Applications of 3D Animation

- **Movies & TV Shows:** Create realistic animated scenes.
- **Video Games:** Design characters, worlds, and special effects.
- **Advertisements:** Make eye-catching visuals for products.
- **Virtual Reality:** Build immersive 3D experiences.
- **Education:** Explain complex topics visually.
- **Science & Engineering:** Simulate realistic models and experiments.

Benefits of 3D Animation

- **Realistic visuals:** Detailed and life-like characters and environments.
- **Better storytelling:** Makes stories more engaging and immersive.
- **Versatile:** Can be used in films, games, ads, education, and more.
- **Interactive:** Enables interactive experiences like VR and video games.

9. Special Effects in Animation

Special effects (SFX) in animation are techniques used to make scenes look more realistic or magical—things that are hard or impossible to do in real life. They make animations more exciting and visually impressive.

Types of Special Effects

1. **Particle Effects:** Small elements like smoke, fire, dust, or water that move realistically.
2. **Lighting Effects:** Using light to create moods, shadows, and a sense of depth.
3. **Fluid Simulations:** Realistic movement of water, cloth, or hair in animations.
4. **Shaders and Textures:** Make surfaces look real, like skin, metal, or fabric.
5. **Compositing:** Combining different layers of animation or live-action footage to make a final scene.
6. **Motion Capture:** Recording real human movements and applying them to animated characters.

Examples of Special Effects

- **Explosions & Fire:** Realistic action scenes with smoke and flames.
- **Magical Effects:** Glowing, sparkling, or magical powers in fantasy scenes.
- **Weather Effects:** Rain, snow, wind, or storms to make scenes more realistic.
- **Character Enhancements:** Special powers, glowing eyes, or transformations.

Importance of Special Effects

- Makes animations more **realistic** or **fantastical**.
- Enhances **visual appeal** and **storytelling**.
- Allows animators to create things that are **impossible in real life**.
- Helps make the animation **memorable and engaging** for viewers.

10. Creating Animation

Animation is a versatile tool that can communicate ideas, explain products/services, or promote brands effectively. Many companies don't fully use animation because of myths like "it's too expensive" or "it takes too long," but these are mostly misconceptions.

The animation production process involves **8 key steps**:

1. Researching

- Understand the **purpose**, **audience**, and **requirements** of the animation project.
- Involve the **whole team** to ensure everyone is on the same page.
- Goal: Gather information to create a solid foundation for the animation.

2. Defining the Project's Scope

- Decide the **message** of the animation.
- Set **timelines, deadlines, and budget**.
- Define ROI and goals.
- Importance: Prevents delays, poor animation quality, and cost overruns.

3. Brainstorming & Script Writing

- Focus on the **central theme** and **script**.
- Key points for animation scripts:
  - **Brevity:** Animations are usually short (e.g., <30 seconds for ads).
  - **Clarity:** Message must be simple and understandable.
  - **Consistency:** Tone of script should match animation style (fun, serious, quirky).
- The script is the **backbone** of the animation.

4. Voiceover Recording

- Add **voiceovers** to match the script.
- Ensure **tone, style, and intonation** complement the animation.
- Voiceovers enhance storytelling and audience engagement.

5. Storyboarding

- A storyboard is a **scene-by-scene visual plan** of the animation.
- Importance:
  - Visualizes the **script and flow** of the animation.
  - Helps spot mistakes and improve transitions.
  - Saves **time and money** during production.

6. Illustration

- Create **characters, objects, and backgrounds** based on the storyboard and script.
- Use **style frames** to test different visual styles.
- Goal: Choose the style that best communicates the message and appeals to the audience.

7. Animation

- Bring everything **to life** using animation software.
- Key aspects:
  - **Natural movements:** Make characters and objects move realistically.
  - **Seamless transitions:** Smoothly connect scenes.
  - **Lighting:** Set mood, focus, and depth.
- High-quality animation depends on **movement, transitions, and lighting**.

8. Sound Design

- Add **music and sound effects** that match the animation's tone and flow.
- Sound is as important as visuals for **engaging the audience**.
- Use **royalty-free resources** if needed, but ensure the sound complements the animation.

11. Creating Animation in Flash

Flash animations can be created in **Adobe Animate** (formerly Flash Professional) and embedded in **PDFs** using Adobe Acrobat Pro. Flash uses **symbols, keyframes, and tweens** to create interactive and smooth animations.

1. Creating Animations in Adobe Animate (Flash)

1. **Start a New Document:** Use the **ActionScript 3.0 template** to create a new project.
2. **Draw or Import Assets:** Create characters, objects, or backgrounds using **drawing tools**, or import images.
3. **Convert to Symbols:**
   - Convert drawings into **symbols** (Graphic or Movie Clip) using **Modify → Convert to Symbol**.
   - Symbols allow easy reuse and animation.
4. **Create Keyframes:** Insert keyframes at different points in the **timeline** to define changes in position, size, rotation, or color.

5. **Apply Tweens:**
    - **Motion Tween:** Smooth movement between keyframes.
    - **Shape Tween:** Morph one shape into another for fluid transformations.
6. **Frame-by-Frame Animation:** For precise control, draw **each frame individually** to create detailed animations.
7. **Test and Refine:** Use **Control → Test Movie** (or Ctrl+Enter) to preview and adjust your animation.
8. **Export as SWF:** Export the animation as a **Flash (.swf) file** using **File → Export → Export Movie**.

2. Embedding Animations in PDF (Adobe Acrobat Pro)

1. **Open PDF:** Open your document in **Adobe Acrobat Pro**.
2. **Rich Media Tools:** Go to **Tools → Rich Media**.
3. **Add Multimedia:** Choose the type of media (e.g., SWF file).
4. **Select SWF File:** Browse and select the exported **.swf file**.
5. **Position and Customize:** Draw a rectangle to place the animation and adjust **playback settings**.
6. **Save PDF:** Save the PDF to embed the animation.

Key Flash Animation Techniques

- **Motion Tweening:** Creates smooth movement between keyframes.
- **Shape Tweening:** Morphs one shape into another.
- **Frame-by-Frame Animation:** Provides full control over every frame for detailed effects.